1578–1657

British physician and natural historian William Harvey was the first person to understand how blood circulates around the body and to correctly explain the role the heart plays in this process. His work replaced centuries-old speculation with a theory based on his research into the circulatory systems of animals.

William Harvey was the son of a wealthy merchant. He studied medicine at Cambridge University and then at the University of Padua, Italy. His teacher at Padua, the esteemed anatomist Hieronymus Fabricius, was a great influence on him. Fabricius had discovered that human veins had one-way valves, although he did not understand why.

On his return to London, Harvey became a lecturer in surgery. His reputation grew, and he was appointed Royal Physician. But his main interest was in research. He was dissatisfied with the prevailing view of circulation, which dated back to the ancient Greek medical practitioner Galen (see pp.24–29), who believed that blood was made in the liver and was sucked from the veins by the heart.

Harvey believed that experimentation was crucial to medicine and carried out many dissections—not only of dead bodies but also of living animals. Although the latter was brutal by modern standards, it led to his discovery that the heart is a pump and that it drives the circulation of the blood. He described this system in detail and accurately predicted the existence of capillaries, which transfer blood from the arteries to the veins, thereby completing its circulation around the body.

MILESTONES

RADICAL THEORY
Tells the Royal College of Physicians, at the 1616 Lumleian lecture, that the heart drives circulation.

ROYAL DOCTOR
Becomes court physician to King James I in 1618, and then to King Charles I in 1632.

SEMINAL TEXT
Publishes *De Motu Cordis*, his text on blood and the heart, in 1628, dedicating it to King Charles I.

“The blood is driven into a round … **and it moves perpetually.**”

William Harvey, 1628

Harvey explained *his theory of circulation in his key work* De Motu Cordis (On the Motion of the Heart). *In it, he illustrated how one-way valves in the veins ensure that blood flows in only one direction.*